Physics

Module 3 : Electrodynamics

Operators ;

1) Gradient = $\vec{\nabla}\phi$

2) Divergence = $\vec{\nabla}\cdot\vec{\phi}$

3) Curl : $\vec{\nabla}\times\vec{\phi}$

$$\vec{\nabla} = \hat{i}\frac{\partial}{\partial x} + \hat{j}\frac{\partial}{\partial y} + \hat{k}\frac{\partial}{\partial z}$$

Charge Distributions

1) Line charge Density

$$\lambda = \frac{dq}{dl} \rightarrow q = \int dq = \int_l \lambda dl$$

2) Surface Charge Density

$$\sigma = \frac{dq}{ds} \rightarrow q = \int dq = \int_s \sigma ds$$

3) Volume Charge density

$$\rho = \frac{dq}{dV} \rightarrow q = \int dq = \int_V \rho dv$$

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* Fundamental Theorems

1) $\int_a^b \vec{\nabla}\phi \, dl = \phi(b) - \phi(a)$

2) Gauss divergence theorem

$$\int_S \vec{E} \cdot \vec{ds} = \int_V (\vec{\nabla} \cdot \vec{E}) dV$$

3) Stoke's theorem

$$\int_L \vec{E} \cdot \vec{dl} = \int_S (\vec{\nabla} \times \vec{E}) \cdot \vec{ds}$$

4) Equation of continuity

$$\vec{\nabla} \cdot \vec{J} + \frac{\partial \rho}{\partial t} = 0$$

→ Maxwell's First Equation;

→ Gauss's law of electrostatics:

The electric field in an enclosed surface is same as the net amount of charges enclosed in the given space

Mathematically,

$$\oint_S \vec{E}\cdot\vec{ds} = \phi_{net} \longrightarrow (1)$$

ϕ_{net} : Net electric flux generated

$$\phi_{net} = \frac{q_{net}}{\epsilon_0} \longrightarrow (2)$$

$$\oint \vec{E}\cdot ds = \frac{1}{\epsilon_0}\cdot q_{net} \longrightarrow (3)$$

As the charges distributed over a surface is calculated as volume charge density

Therefore, $\oint \vec{E}\cdot\vec{ds} = \frac{1}{\epsilon_0}\int_V \rho dV \longrightarrow (4)$

Integral form

Eqⁿ (4) is the integral form of Maxwell's first eqⁿ.

Now, using Gauss's divergence theorem, eqⁿ (4) can be modified as

$$\oint_S \vec{E}\cdot\vec{ds} = \int_V (\vec{\nabla}\cdot\vec{E})\,d\vec{V} \longrightarrow (5)$$

$$\int (\vec{\nabla}\cdot\vec{E})dV = \frac{1}{\epsilon_0}\int_V \rho dV$$

The volume integral nullifies on both the sides

$$\vec{\nabla}\cdot\vec{E} = \rho/\epsilon_0 \longrightarrow \text{Differential form}$$

As $D = \epsilon_0 E$,

$$\oint_S \vec{D}\,\vec{ds} = \oint_V \rho dV \quad \text{or} \quad \vec{\nabla}\cdot\vec{D} = \rho$$

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Maxwell's Second Equation

→ **Gauss's law of Magnetostatics**

The magnetic field in an enclosed surface is same as the net amount of magnetic lines of forces in the given region.

Magnetic flux density,

$$B = \frac{\phi}{A} \longrightarrow (1)$$

In Integral form, the eq^n (1) can be expressed as

$$\oint \vec{B} \cdot \vec{ds} = \phi_{net} \longrightarrow (2)$$

As the magnetic lines of forces when enters a surface, they will surely exit. Hence, the net flux is zero.

→ Eq^n (2) now becomes,

$$\boxed{\oint_s \vec{B} \cdot \vec{ds} = 0} \rightarrow (3)$$

Integral form of Maxwell's 2nd eq^n

Now, applying Gauss's divergence theorem, to (eq^n) 3

$$\rightarrow \oint_S \vec{B} \cdot \vec{ds} = \oint_V (\vec{\nabla} \cdot \vec{B})\, dV \longrightarrow (4)$$

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→ eq^n (3) becomes

$$\int (\vec{\nabla} \cdot \vec{B})\, dV = 0$$

$$\boxed{\therefore\ \vec{\nabla} \cdot \vec{B} = 0}$$ → Differential eq^n

↓

Magnetic monopoles never exist

** Possible question **

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Maxwell's Third Equation

→ **Faraday's Law of EMI**

The current applied to the primary coil can induce an emf in a secondary connected coil in a closed loop.

Mathematically emf $e = -\frac{d\phi}{dt} \longrightarrow (1)$

$e = -\frac{d}{dt}(B \cdot ds)$ As $B = \phi/A$

$\hookrightarrow (2)$

As the emf is induced in the closed loop

$$e = \oint_L \vec{E} \cdot \vec{dl} \longrightarrow (3)$$

we compare both the equations (2) & (3)

$$\rightarrow \int_l \vec{E} \cdot \vec{dl} = \int -\frac{d}{dt}(\vec{B} \cdot ds) \longrightarrow (4)$$

$$\boxed{\oint_l \vec{E} \cdot \vec{dl} = -\int \frac{\partial \vec{B}}{\partial t} \cdot \vec{ds}} \longrightarrow (5)$$

Integral form of maxwell's 3^{rd} eq^n

Apply Stoke's Theorem to eq^n (5)

$$\oint_l \vec{E} \cdot \vec{dl} = \oint_s (\vec{\nabla} \times \vec{E}) \cdot \vec{ds} \rightarrow (6)$$

Replace LHS of eq^n (5) with (6)

$$\rightarrow \oint_s (\vec{\nabla} \times \vec{E}) \vec{ds} = -\int \frac{\partial B}{\partial t} \cdot \vec{ds} \rightarrow (7)$$

$$\boxed{\rightarrow \vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}} \rightarrow (8)$$

Differential form of maxwell's 3rd eq^n

Maxwell's Fourth Equation

→ Ampere's circuital law

$$\oint \vec{B} \cdot \vec{dl} = \mu_0 I \rightarrow (1)$$

The magnetic field along a closed path is equal to μ_0 times the net current flowing along the path.

$$\oint \vec{B} \vec{dl} = \mu_0 \int \vec{J} \cdot \vec{ds} \rightarrow (2)$$

As $J = \frac{I}{A}$

Apply Stoke's theorem to eq^n (2)

$$\rightarrow \oint_l \vec{B}\, d\vec{l} = \oint_s (\vec{\nabla} \times \vec{B}) \cdot \vec{ds} \rightarrow (3)$$

→ eq^n (2) can be written as

$$\oint_s (\vec{\nabla} \times \vec{B}) \cdot \vec{ds} = \mu_0 \oint_s \vec{J} \cdot \vec{ds} \rightarrow (4)$$

$$\rightarrow \boxed{\vec{\nabla} \times \vec{B} = \mu_0 \vec{J}} \rightarrow (5)$$

Take divergence of eq^n (5)

$$\vec{\nabla} \cdot (\vec{\nabla} \times \vec{B}) = \mu_0 (\vec{\nabla} \cdot \vec{J}) \rightarrow (6)$$

Divergence of a curl is 0

$$\rightarrow \vec{\nabla} \cdot (\vec{\nabla} \times \vec{B}) = 0$$

$$\rightarrow \vec{\nabla} \cdot \vec{J} = 0 \rightarrow (7)$$

↳ Incorrect

The equation was modified by Ampere by introducing Eq^n of continuity

$$\vec{\nabla} \cdot \vec{J} + \frac{\partial \rho}{\partial t} = 0 \rightarrow (8)$$

Using first maxwell's eq^n $\vec{\nabla} \cdot \vec{E} = \rho / \epsilon_0 \rightarrow (9)$

$$\rightarrow \rho = \vec{\nabla} \cdot \epsilon_0 \vec{E} \rightarrow (10)$$

Substitute (10) in (8)

$$\rightarrow \vec{\nabla}.\vec{J} + \frac{\partial}{\partial t}(\vec{\nabla}.\epsilon_0\vec{E}) = 0$$

$$\rightarrow \vec{\nabla}.\vec{J} + \vec{\nabla}.\left(\frac{\partial}{\partial t}(\epsilon_0\vec{E})\right) = 0$$

$$\rightarrow \vec{\nabla}.\left[\vec{J} + \epsilon_0\frac{\partial E}{\partial t}\right] = 0 \longrightarrow (11)$$

$\vec{J}$ → Steady state current

$\epsilon_0 \frac{\partial E}{\partial t}$ → contribution of displacement current or time varying E.F.

Substitute eq^n (11) in eq^n (5)

$$\rightarrow \vec{\nabla}\times\vec{B} = \mu_0\left[\vec{J} + \epsilon_0\frac{\partial \vec{E}}{\partial t}\right] \rightarrow (12)$$

$$\rightarrow \vec{\nabla}\times\vec{B} = \mu_0\left[\vec{J} + \frac{\partial \vec{D}}{\partial t}\right] \rightarrow (13)$$

(12), (13): <u>Differential form of Maxwell's 4th eq^n</u>

$$\oint_l \vec{B}.\vec{dl} = \mu_0\oint_s\left(\vec{J} + \epsilon_0\frac{\partial E}{\partial t}\right)\vec{ds} \rightarrow (14)$$

As $B = \mu_0 H$,

$$\rightarrow \oint_l \vec{H}.\vec{dl} = \oint_s\left(\vec{J} + \epsilon_0\frac{\partial E}{\partial t}\right)\vec{ds} \rightarrow (15)$$

(15) & (14) <u>Are the Integral forms of Maxwell's 4th Equation</u>

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Electromagnetic Wave Equation

The 4th Maxwell's equation can be written as (differentiated form)

1) $\vec{\nabla}\cdot\vec{E} = \rho/\varepsilon_0 \rightarrow (1)$

2) $\vec{\nabla}\cdot\vec{B} = 0 \rightarrow (2)$

3) $\vec{\nabla}\times\vec{E} = \frac{-\partial\vec{B}}{\partial t} \rightarrow (3)$

4) $\vec{\nabla}\times\vec{B} = \mu_0\left[\vec{J} + \varepsilon_0\frac{\partial\vec{E}}{\partial t}\right] \rightarrow (4)$

In free space,

$\rho = 0$ & $J = 0$

$\therefore$ eq^n (1) & eq^n (4) can be written as

$\vec{\nabla}\cdot\vec{E} = 0 \rightarrow (5)$

$\vec{\nabla}\times\vec{B} = \mu_0\left[\varepsilon_0\frac{\partial\vec{E}}{\partial t}\right] \rightarrow (6)$

Take curl of eq^n (3)

$\therefore \vec{\nabla}\times(\vec{\nabla}\times\vec{E}) = \vec{\nabla}\times\left(\frac{-\partial\vec{B}}{\partial t}\right)$

$\rightarrow \vec{\nabla}\times(\vec{\nabla}\times\vec{E}) = \frac{-\partial}{\partial t}(\vec{\nabla}\times\vec{B}) \rightarrow (7)$

Now, on LHS

$$\vec{\nabla} \times (\vec{\nabla} \times \vec{E}) = \vec{\nabla}(\vec{\nabla} \cdot \vec{E}) - \nabla^2 E \rightarrow (8)$$

As $\vec{\nabla} \cdot \vec{E} = 0$

$$\rightarrow \vec{\nabla} \times (\vec{\nabla} \times \vec{E}) = -\nabla^2 E \rightarrow (9)$$

Substitute (9) & (6) in (7)

$$\rightarrow -\nabla^2 E = \frac{-\partial}{\partial t}\left(\mu_0 \left[\epsilon_0 \frac{\partial \vec{E}}{\partial t}\right]\right)$$

$$\rightarrow +\nabla^2 E = +\mu_0 \epsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2} \rightarrow (10)$$

One-D wave eq^n can be written as;

$$\frac{\partial^2 y}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 y}{\partial t^2} \rightarrow (11)$$

If (10) & (11) are compared, we get

$$\frac{1}{V^2} = \mu_0 \epsilon_0 \rightarrow V = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$ Velocity of an EM wave

$\epsilon_0 = 8.85 \times 10^{-12}$
$\mu_0 = 4\pi \times 10^{-7}$

$$\rightarrow V = \frac{1}{\sqrt{8.85 \times 10^{-12} \times 4\pi \times 10^{-7}}}$$

$$= 3 \times 10^8$$

Speed of Light